Mingsi Liao

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Research Interests

Artificial Intelligence (Machine learning / Deep learning / Computer vision)
Genomics and Sequencing Technologies (Next-generation sequencing / Metatranscriptomics / Whole-genome sequencing)

Education

Virginia Polytechnic Institute and State University — Blacksburg, VA, USA
Doctor of Philosophy (Ph.D.) — 2022 -- Present

- Genetics, Bioinformatics and Computational Biology
- School of Animal Sciences
- Dissertation: Integration of early dietary supplementation and automated feeding systems to mitigate postweaning slump in dairy heifers & Use of precision technology to predict pathogenic diarrhea in pre-weaned dairy heifers.
- Advisors: Dr. Rebecca Cockrum (Chair), Dr. Chris Thomas (Co-Chair), Dr. Gota Morota, Dr. Christina Petersson-Wolfe

Virginia Polytechnic Institute and State University — Blacksburg, VA, USA
Master of Engineering (M.Eng.) — 2025 -- 2026

- Computer Science and Applications
- Department of Computer Science
- Concentration Area: Data Analytics and Artificial Intelligence

George Mason University — Fairfax, VA, USA
Master of Science (M.S.) — 2019 -- 2021

- Bioinformatics and Computational Biology
- School of Systems Biology
- Project and Report (Non-thesis): Genomic and bioinformatic analyses of an ocular pathogen that molecularly types as a respiratory pathogen
- Advisor: Dr. Don Seto

University of Electronic Science and Technology of China, Zhongshan Institute — Zhongshan, Guangdong, China
Bachelor of Science (B.S.) — 2013 -- 2017

- Biotechnology
- School of Materials Science and Food Engineering
- Exchange Student, Chinese Culture University, Taipei, Taiwan (2015 -- 2016)
- Capstone Project: Exploitation and development of natural pigments from black olives
- Advisor: Dr. Lin Li

Research Projects and Publications

AI and Computer Vision for Automated Calf Body Weight Prediction

- Developed automated data collection system for depth image acquisition
- Implemented CNN-based image segmentation for body dimension extraction
- Conducted ML modeling with cross-validation and longitudinal analysis for weight prediction
- Technical Skills: Deep Learning/Machine Learning, Computer Vision, Automation (System/Camera)
- **Publication:**
  - **Liao, M.**, Morota, G., Bi, Y., & Cockrum, R. (2025). Predicting Dairy Calf Body Weight from Depth Images Using Deep Learning (YOLOv8) and Threshold Segmentation with Cross-Validation and Longitudinal Analysis. *Animals*, 15(6), 868.

Whole Genome Sequencing for Breed Differentiation in Sheep

- Performed variant discovery and filtering from whole-genome sequencing data
- Conducted population genetic analyses to identify breed-specific signatures
- Integrated functional annotation and QTL data for trait associations
- Technical Skills: Whole Genome Sequencing, Variant Calling, Population Genetics, Bioinformatics
- **Publication:**
  - **Liao, M.**, Kravitz, A., Haak, D., R., Cockrum. and Sriranganathan, N., (2026). Whole-Genome Sequencing Reveals Breed-Specific SNPs, Indels, and Signatures of Selection in Royal White and White Dorper Sheep. *Animals* 16(5), 811.

Genetic Association Analysis in Ovine Johne's Disease

- Performed GWAS to identify SNPs linked to disease resistance using SNP50 BeadChip data
- Applied QC, genotype imputation, and logistic regression for variant discovery
- Technical Skills: GWAS, Genotype Imputation, Logistic Regression, Bioinformatics Pipelines
- **Publication:**
  - Kravitz, A., **Liao, M.**, Morota, G., et al. (2024). Retrospective SNP Analysis of Host Resistance to Ovine Johne's Disease. *Int. J. Mol. Sci.*, 25(14), 7748.

Deep Learning for Diarrhea Detection in Dairy Calves

- Trained and fine-tuned deep learning models for image-based diarrhea classification
- Benchmarked vision-language models under zero-shot and few-shot settings
- Built end-to-end pipelines for image preprocessing, training, and evaluation
- **Technical Skills**: Computer Vision, Deep Learning, Vision-Language Models, Python (PyTorch)
- **Publication (Under Review):**
  - **Liao, M.**, Atabuzzaman, M., Thomas, C., & Cockrum, R. Dairy Calf Diarrhea Detection Using Real-World Farm Data: A Comparative Study of Deep Learning and Vision-Language Models.

Early Lactose Supplementation and Calf Microbiome Development

- Conducted longitudinal 16S rRNA microbiome analyses in dairy calves
- Integrated microbiome, growth, health, and feeding data for statistical modeling
- Performed diversity, community composition, and differential abundance analyses
- **Technical Skills**: Microbiome Analysis, 16S rRNA Sequencing, QIIME2, R, Statistical Modeling
- **Publication (In Preparation):**
  - **Liao, M.**, Bowers, O., & Cockrum, R. Early Lactose Supplementation and Longitudinal Microbiome Development in Dairy Calves. In preparation.

Multimodal Prediction of Weaning Readiness and Diarrhea in Dairy Calves

- Integrated depth imaging, behavioral, health, and microbiome data for multimodal learning
- Developed predictive models for weaning readiness and disease forecasting in dairy calves
- Built longitudinal data pipelines for feature extraction, fusion, and model development
- **Technical Skills**: Machine Learning, Computer Vision, Multimodal Learning, Time-Series Analysis, PyTorch
- **Status**: Ongoing research project

Professional Skills & Certificates

Computational Skills: Machine Learning/Deep Learning (CNNs, transformer, transfer learning, regression, clustering), Computer Vision (object detection, segmentation), Vision Language and Multimodal Models (Qwen VL, grounding based detection, zero shot and few shot inference), Python (data analysis, PyTorch, scikit-learn), R (statistical modeling, visualization)

Bioinformatics & Genomic Analysis: Next-generation sequencing (quality control, alignment, variant calling/annotation), GWAS (imputation, association, population genetics)

Soft Skills: Leadership (mentored and coordinated 20 undergraduate students in research projects), grant writing, teamwork, problem-solving, adaptability

Certifications: Graduate Certificate in Data Analytics, Institutional Animal Care and Use Committee (IACUC) Training Certificate, USA

Honors and Awards

2025	**National Research Support Project (NRSP-8) fellowship grant, USA**
2025	**Center for Advanced Innovation in Agriculture (CALS) travel award, USA**
2024	**2nd Place Award in Poster Presentation (GPSS Research Symposium), USA**
2024	**Virginia Tech School of Animal Sciences Research Symposium Travel Award, USA**
2023	**Virginia Tech School of Animal Sciences Research Symposium Travel Award, USA**
22-25	**John Lee Pratt Animal Nutrition Scholarship, USA**
2016	**2nd Place Institutional Scholarship, China**
2015	**1st Place Institutional Scholarship, China**
2014	**1st Place Institutional Scholarship, China**

Experience

Virginia Polytechnic Institute and State University Blacksburg, VA
Research Assistant 2022 -- Present

- Designed and developed research projects in collaboration with principal investigators
- Conducted literature reviews, data collection, and analysis for research objectives
- Prepared IACUC protocols ensuring compliance with regulatory and ethical standards
- Implemented data management protocols and maintained accurate experimental records
- Participated in team meetings to brainstorm, troubleshoot, and evaluate outcomes
- Disseminated findings through presentations, reports, and scientific publications
- Collaborated with interdisciplinary teams to facilitate research and enhance outcomes

Virginia Polytechnic Institute and State University Blacksburg, VA
Project Team Leader 2023 -- 2025

- Led and supervised a team of undergraduate researchers for projects
- Oversaw recruitment, task assignment, and training initiatives
- Facilitated communication and collaboration within the team

Virginia Polytechnic Institute and State University Blacksburg, VA
Teaching Assistant Fall 2022 / Fall 2023 / Fall 2024 / Fall 2025

- Course: Animal Breeding and Genetics 3104
- Managed undergraduate teaching assistants and scheduled office hours.
- Addressed student questions related to coursework and assignments.
- Assisted the instructor with grading exams.

George Mason University Fairfax, VA
Graduate Researcher 2020 -- 2021

- Project: Genomic and Bioinformatic Analyses of an Ocular Pathogen that Molecularly Types as a Respiratory Pathogen
- Analyzed a novel pathogen using bioinformatics workflows: BLAST, annotation, phylogenetic analysis, genome comparison, and recombination analysis
- Identified recombination events from known human adenoviruses, contributing to pathogen classification insights

University of Electronic Science and Technology of China, Zhongshan Institute Zhongshan, Guangdong, China
Undergraduate Researcher 2016 -- 2017

- Capstone Project: Exploitation and Development of Natural Pigments from Black Olives
- Extracted polyphenols and anthocyanins from olive skin and pulp; optimized extraction conditions
- Assessed pigment stability under heat, oxidation, and reduction; evaluated effects of additives and metal ions

Presentations

Genetics, Bioinformatics and Computational Biology Seminar, Virginia Tech Blacksburg, VA
Whole-Genome Sequencing Reveals Breed-Specific SNPs, Indels, and Signatures of Selection in Royal White and White Dorper Sheep (Oral Presentation) 2026

The Plant and Animal Genome Conference (PAG) San Diego, CA
Vision-Based Monitoring of Diarrhea in Pre-Weaning Dairy Calves (Invited Talk) 2026

Reproductive Biology Club, School of Animal Science, Virginia Tech Blacksburg, VA
Breed-Specific Genetic Variants and Selective Sweep Regions in Royal White and White Dorper Sheep Revealed Through Whole-Genome Sequencing (Oral Presentation) 2025

American Society of Animal Science (ASAS) Conference Hollywood, FL
Genome-Wide Characterization of SNPs and Indels in Royal White and White Dorper Sheep Using Whole-Genome Sequencing (Oral Presentation) 2025

Research Day, School of Animal Science, Virginia Tech Blacksburg, VA
Genome-Wide Characterization of SNPs and Indels in Royal White and White Dorper Sheep Using Whole-Genome Sequencing (Poster Presentation) 2025

Graduate and Professional Student Senate (GPSS) 41st Annual Research Symposium and Exposition Blacksburg, VA
Predicting Dairy Calf Body Weight from Depth Images Using Deep Learning (YOLOv8) and Threshold Segmentation with Cross-Validation and Longitudinal Analysis (Poster Presentation) 2025

Genetics, Bioinformatics and Computational Biology Seminar, Virginia Tech Blacksburg, VA
Predicting Dairy Calf Body Weight from Depth Images Using Deep Learning and Threshold Segmentation with Cross-Validation and Longitudinal Analysis (Oral Presentation) 2025

Center for Advanced Innovation in Agriculture Big Event Blacksburg, VA
Predicting Dairy Calf Body Weight from Depth Images Using Deep Learning and Threshold Segmentation with Cross-Validation and Longitudinal Analysis (Poster Presentation) 2025

The Plant and Animal Genome Conference (PAG) San Diego, CA
Comparison of Deep Learning and Threshold-Based Methods for Depth Image Segmentation Using Extreme Gradient Boosting to Predict Dairy Calf Body Weight (Poster Presentation) 2025

American Dairy Science Association (ADSA) West Palm Beach, FL
Dietary Crude Protein is Independent of Rumen Microbial Community Ecology in Lactating Dairy Cows (Poster Presentation) 2024

American Dairy Science Association (ADSA) West Palm Beach, FL
Dairy Calf Health and Growth Monitoring Through Camera Phenotyping Techniques (Oral Presentation) 2024

Research Day, School of Animal Science, Virginia Tech Blacksburg, VA
Camera-Based Monitoring of Dairy Calf Health and Growth: Phenotyping Techniques Applied (Oral Presentation) 2024

Dairy Science Seminar, Virginia Tech Blacksburg, VA
Colostrum Quality: Management and Storage for Healthy Calves (Oral Presentation) 2024

40th Annual Graduate and Professional Student Senate (GPSS) Research Symposium Blacksburg, VA
Dietary Crude Protein is Independent of Rumen Microbial Community Ecology in Lactating Dairy Cows (Poster Presentation) 2024

Center for Advanced Innovation in Agriculture Big Event Blacksburg, VA
Dairy Calf Health and Growth Monitoring Through Camera Phenotyping Techniques (Oral Presentation) 2024

Reproductive Biology Club, School of Animal Science, Virginia Tech Blacksburg, VA
Work in Progress: Dairy Calf Health and Performance -- Research Impact and Future Direction (Oral Presentation) 2024

Reproductive Biology Club, School of Animal Science, Virginia Tech Blacksburg, VA
Work in Progress: Unveiling the Mysteries of the Dairy Cow Rumen: Microbial Composition and Function (Oral Presentation) 2023

Research Day, School of Animal Science, Virginia Tech Blacksburg, VA
Shotgun Metatranscriptomics Reveals Functional Profiles of Rumen Microbial Ecosystem in Dairy Cows (Oral Presentation) 2023

Graduate Coursework

Ph.D. & M.Eng. (Virginia Tech):

- CS 5024 Ethics & Professionalism in Computer Science
- CS 5040 Intermediate Data Structures & Algorithms
- CS 5525 Data Analytics
- CS 5664 Social Media Analytics
- CS 5806 Machine Learning II
- CS 5814 Introduction to Deep Learning
- CS 5914 TS: Grad Course Design in CS
- CS 5934 Capstone Project
- CS 5984 Special Studies: Engineering Entrepreneurship
- ECE 5554 Computer Vision
- STAT 5615 Statistics in Research
- STS 5444 Issues in Bioethics
- GBCB 5874 Problem Solving in Genetics, Bioinformatics, and Computational Biology

M.S. (George Mason University):

- BINF 630 Bioinformatics Methods
- BINF 631 Molecular Cell Biology
- BINF 633 Molecular Biotechnology
- BINF 634 Bioinformatics Programming
- BINF 701 Systems Biology
- BINF 702 Biological Data Analysis
- BINF 731 Protein Structure Analysis
- BINF 739 Next-Generation Sequencing
- BINF 739 Computational Analysis of Viral Genomes
- BIOL 695 Coronavirus Research Update